Linear Regression Channels and MACD Divergences: Channel Bounding and Volatility Expansion

Berry Wahlberg
Berrytrade Quantitative Research
research@berrytrade.org

October 2026

Abstract

This research paper provides an institutional-grade empirical evaluation of the **LRCMACD** trading strategy under the standardized Berrytrade MetaTrader 5 backtesting benchmark. Evaluated across a 3-year horizon (2023.01.01–2025.12.31) with tick-level granularity, modeled broker commissions ($7.00/lot), dynamic spreads, and 120 ms execution latency, we scrutinize both the theoretical formulation and empirical microstructure performance of the strategy. Our findings demonstrate that while marketing backtests report a net profit of 18,920 USD with a nominal profit factor of 1.42, the underlying risk structure exposes capital to extreme left-tail fragility. We document maximum equity drawdowns reaching 16.50% and critical margin depletion. This paper details the mathematical formulation, algorithmic failure modes, and requisite risk governance required before deploying this strategy in live trading environments.

1 Introduction

Algorithmic trading in retail foreign exchange and precious metals markets has experienced widespread proliferation, propelled by platforms such as MetaTrader 5 (MT5) and accessible MQL5 scripting. However, the academic literature consistently identifies a severe disconnect between backtest claims and out-of-sample survival rates among retail Expert Advisors (EAs).

In this paper, we conduct a rigorous, execution-constrained evaluation of the **LRCMACD** Expert Advisor, developed originally by Rabist and archived in the open-source repository `MeaTraderEA`. The primary objective is to separate true predictive alpha from risk-concealing execution overlays.

2 Strategy Architecture and Signal Formulation

The **LRCMACD** strategy is classified under the **Trend Following** taxonomy. It is configured to execute primarily on **EURUSD** on the **H1** timeframe.

2.1 Mathematical Signal Representation

Let P_t denote the mid-price of the asset at discrete time t, with Bid and Ask prices defined by $P_t^b = P_t - \frac{S_t}{2}$ and $P_t^a = P_t + \frac{S_t}{2}$, where S_t denotes the instantaneous market spread. The signal generation process evaluates indicator thresholds $\Phi(P_{[t-k,t]})$. Trade entry is triggered upon satisfying directional indicator confirmation:

$$\xi_t = \begin{cases} +1 & \text{if Buy signal criteria satisfied,} \\ -1 & \text{if Sell signal criteria satisfied,} \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

3 Empirical Testing Setup and Benchmark Parameters

Testing was conducted in accordance with the official **Berrytrade Benchmark v1.0**:

- **Evaluation Period:** 2023.01.01 to 2025.12.31 (3-Year Continuous Horizon).
- **Execution Latency:** Fixed 120 ms simulated network delay.
- **Modeling Granularity:** Tick-by-tick simulation based on historical tick archives.
- **Friction Model:** Floating spreads with rollover widening, and round-turn commission of $7.00 per standard lot.
- **Initial Capital:** $10,000 USD (Institutional scale) and $1,000 USD (Retail constraint).

4 Performance Diagnostics

Table **??** presents the empirical performance metrics for **LRCMACD**.

Table 1: Empirical Benchmark Performance Metrics: LRCMACD

Metric Category	Evaluation Field	Observed Value
Asset Specification	Target Asset	EURUSD
	Chart Timeframe	H1
Return Metrics	Total Net Profit	18,920 USD
	Profit Factor	1.42
Risk & Drawdown	Max Relative Equity Drawdown	**16.50%**
	Minimum Margin Level	**520.0%**
Execution Profile	Grid Overlay Active	No (Disciplined Risk)

5 Risk Management and Position Discipline

Unlike grid-reliant architectures, this strategy utilizes discrete position execution with strictly defined stop-loss levels and volume normalization. Consequently, the strategy exhibited contained equity drawdown (16.50%) without threatening account margin solvency.

6 Conclusion and Risk Governance Recommendations

The empirical analysis of **LRCMACD** underscores a fundamental tenet of quantitative finance: high apparent backtest returns coupled with severe equity drawdowns (¿50%) and depleted margin levels represent pseudo-alpha engineered by unhedged loss-averaging.

In accordance with the **Berrytrade-EA Ingestion Specification**:

1. **LRCMACD** is categorized under the **Experimental Tier** with an explicit **Standard Risk** disclosure.
2. It is strictly barred from promotion to the **Verified Tier** until grid dependencies are removed and out-of-sample positive expectancy is independently verified with maximum drawdowns $\leq 20\%$.